

## FINAL EXPLANATION: BIOLOGY GRADE 10

### *Student Assessment Document*

#### FINAL EXPLANATION: BIOLOGY GRADE 10: CHEMICALS OF LIFE

##### Student Assessment Document

|              |       |
|--------------|-------|
| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

#### INSTRUCTIONS FOR STUDENTS

You have finished all 6 lessons of the Chemicals of Life unit. Now write your own Final Explanation of Wanjiru's blood glucose graph and answer the driving question: What chemicals are in the food we eat, and what happens to them in our bodies? Use your Summary Tables, your first and final flow-diagram models, your Driving Question Board and the evidence from the class investigations. Write in full sentences and use numbers from the data table in your answers. Use these terms correctly: blood glucose concentration (mmol/L), starch, reducing sugar, digestion, absorption, enzyme, amylase, insulin, respiration and glycogen. Here is Wanjiru's data. After fasting, Wanjiru ate a meal of ugali (300 g), sukuma wiki (100 g) and beans (100 g) between 0 and 15 minutes. Her blood glucose was then measured for 3 hours.

| Time after meal start (min) | Blood glucose concentration (mmol/L) |
|-----------------------------|--------------------------------------|
| 0                           | 4.5                                  |
| 15                          | 5.5                                  |
| 30                          | 7.0                                  |
| 60                          | 8.0                                  |
| 90                          | 6.5                                  |
| 120                         | 5.2                                  |
| 150                         | 4.7                                  |
| 180                         | 4.5                                  |

The normal fasting range is 4.0 to 5.5 mmol/L.

### Part 1: What chemicals are in Wanjiru's lunch?

Wanjiru's lunch was ugali, sukuma wiki and beans. (a) Name the groups of chemicals found in this meal and say which are organic and which are inorganic. State the rule you use to decide. (b) Give the approximate amount of each nutrient in the whole meal. (c) Explain which group of chemicals you think is most likely to cause the rise in her blood glucose concentration, and why this is only a prediction at this stage.

*Write your answer here:*

## Part 2: Food tests as evidence

**(a) Describe the iodine test and Benedict's test: what each one detects and what colour results show a positive result. (b) Describe the biuret test and the grease-spot and emulsion tests in the same way. (c) Give the results you recorded for ugali in all of these tests. (d) Ugali gave no change with Benedict's solution before digestion. What does this result, together with what you know about ugali and the glucose in Wanjiru's blood, tell you about what must happen to the starch before glucose can enter the blood?**

*Write your answer here:*

### Part 3: Enzymes and digestion

**(a) Describe the saliva and starch investigation: what was mixed, the temperature, the tests used and the control. (b) Give the results at 0, 5 and 10 minutes and for the control. (c) Explain these results using the words enzyme, amylase, digestion and reducing sugar. (d) Explain what temperature does to amylase and why 37 degrees C matters. (e) Say what role water and mineral salts play.**

*Write your answer here:*

#### Part 4: Explaining the rise in blood glucose

Use the data table. (a) Calculate the rise in blood glucose concentration from 0 to 60 minutes, and the percentage it is above the starting level. (b) Explain, step by step, why the concentration rose after Wanjiru's meal, using the words digestion and absorption. (c) Explain why you think carbohydrates, and not proteins or lipids, are the main cause of the spike. Use evidence from the meal and from the tests.

*Write your answer here:*

### Part 5: Explaining the fall and the whole story

(a) Calculate the fall in blood glucose concentration from 60 to 180 minutes and compare how fast the rise and the fall happened. (b) State when the level was first back in the normal fasting range. (c) Explain what the body does to bring blood glucose back down, naming the organ and hormone involved and what cells, the liver and muscles do with glucose. (d) Write the equation for respiration in words. (e) Finish by writing a short answer to the driving question: What chemicals are in the food we eat, and what happens to them in our bodies?

*Write your answer here:*

### RUBRIC

| Criterion                                          | Excellent (4)                                                                                                                                                                                 | Proficient (3)                                                                                                 | Developing (1–2)                                                                              |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Chemicals of life (composition and classification) | Correctly names all the food chemical groups, classifies them as organic or inorganic using the carbon-bonded-to-hydrogen rule, names the building blocks and links them to the meal amounts. | Names most chemical groups and classifies them correctly, with some building blocks or meal amounts missing.   | Names few groups, or confuses organic and inorganic, with little reference to the meal.       |
| Use of evidence from investigations                | Uses specific test results (colours, foods, times, mmol/L values) accurately from several lessons to support claims, including the ugali Benedict's result.                                   | Uses some specific results accurately but misses some or states them in general terms.                         | Uses little evidence, or results are vague or incorrect.                                      |
| Enzymes and digestion                              | Explains amylase as a protein catalyst digesting starch to maltose then glucose, uses the saliva results and the boiled-saliva control, and explains the effect of                            | Explains the role of amylase and the main results, with incomplete reasoning about temperature or the control. | Mentions enzymes but does not explain how they digest starch or how temperature affects them. |

|                                           |                                                                                                                                                                                              |                                                                                      |                                                                                   |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|                                           | temperature (37 degrees C, denaturation, inactive at 0 degrees C).                                                                                                                           |                                                                                      |                                                                                   |
| Explaining the phenomenon (rise and fall) | Gives a complete sequence: digestion, absorption, rise to 8.0 mmol/L at 60 min, insulin, uptake, respiration and glycogen storage, fall to 4.5 mmol/L at 180 min, with correct calculations. | Explains both the rise and the fall with minor gaps in the sequence or calculations. | Explains only the rise or only the fall, or has major gaps or calculation errors. |
| Scientific reasoning                      | Links claims to evidence and gives reasons why carbohydrates, not proteins or lipids, are the main cause of the spike; writes clearly using the correct terms.                               | Gives some reasoning linking evidence to claims, with some unsupported statements.   | Mostly lists facts without reasoning or evidence.                                 |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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